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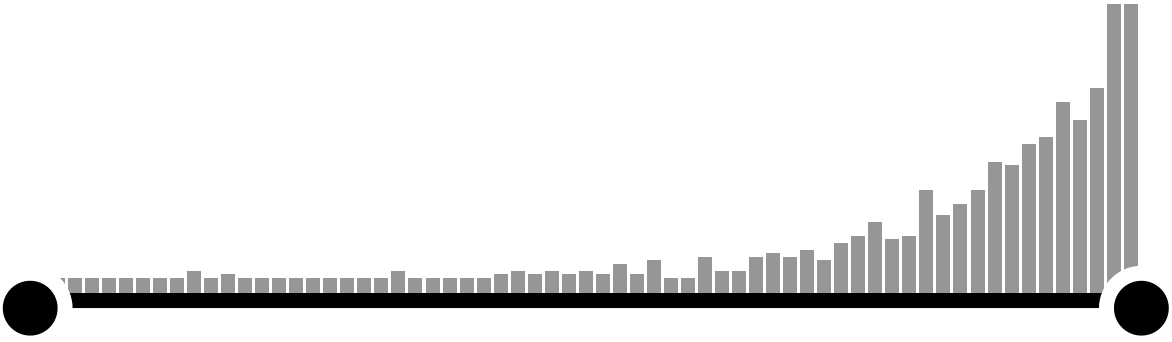
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	Document title	Authors	Source	Year	Citations
601	Conference Paper **Linking daylight and views to the reduction of sick building syndrome**	Heschong L.,Roberts J.E.	9th International Conference and Exhibition Healthy Buildings 2009 Hb 2009	2009	0

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In this paper, the authors explore the hypothesis that the low illumination levels and restricted access to *outdoor* views, commonly experienced in American office buildings and other interior work environments, contribute to circadian dysfunction that can exacerbate occupants' Sick Building Syndrome (SBS) symptoms. If this hypothesis is valid, exposure to daylight illumination and views of the outdoors should logically be included in studies of SBS to see if they are indeed a contributing, or confounding, factor. This paper will first explain recent findings in circadian stimulus and the biochemical linkages to occupant health and well-being, and then discuss findings from a recent study that suggests the luminous environment may be an important factor contributing to seeming unrelated aspects of occupant comfort, health and well being. A field study of 200 office workers analyzed measured and self-reported data about the environmental conditions of occupants' workspaces, their cognitive performance, recent health, and overall satisfaction with their environment. Better access to views was associated with fewer complaints of fatigue, headache, difficulty concentrating and other health complaints, along with fewer complaints about environmental comfort conditions in the building, such as air quality, *thermal* and *acoustic* conditions.

	Document title	Authors	Source	Year	Citations
602	Conference Paper **Development of a micro-satellite compatible FTS sounder for sun-occultation measurements**	Giaccari P.,Moreau L.M., Giroux J.G.,Soucy M.-A.	Proceedings of SPIE the International Society for Optical Engineering , 7474, 74741N	2009	0

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The SciSat/ACE mission provided, and still provides, high quality and high spectral resolution measurements of the atmosphere with a FTS sounder in sun-occultation configuration. Based on the comprehensive results and models of SciSat/ACE it is foreseen that most of the desired information can also be retrieved from lower spectral resolution measurements with higher signal-to-*noise* ratio (SNR) and appropriate data treatment. With the Canadian Space Agency under the Space Technologies Development Program, ABB Analytical developed a small size sun-occultation sounder compatible with a micro-satellite platform that has identical throughput, spectral bandwidth and vertical resolution as ACE. The spectral resolution is decreased by a factor 25 (0.6 cm^{-1} instead of 0.024 cm^{-1} for ACE) whereas the SNR performance is highly increased with an equal factor (target of 2500 instead of 100 for ACE over most of the spectral bandwidth between 750 and 4000 cm^{-1}).A prototype of the sun-occultation sounder was built, tested under various *thermal* conditions and subjected to vibrations similar to those expected at launch. An *outdoor* experiment was also conducted to test the instrument in sun-occultation conditions. The good behavior of the instrument

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indicates interesting opportunities for such small footprint sounder on a low-cost micro-satellite mission and potentially good earth coverage if several of such instruments are used in coordination. Depending on the scientific needs, it is possible to adapt the proposed instrument to increase the vertical resolution and/or to extend the measurements on lower altitudes due to the higher SNR performances. © 2009 SPIE.

Conference Paper

603 **Channel models for optical wireless communication systems** — Joshi H.,Green R.J., Leeson M.S. — Icton 2009 11th International Conference on Transparent Optical Networks, 5184992 — 2009 — 7

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We present an up to date review of various channel models used for optical wireless communication systems so far. Various channel models for both indoor and *outdoor* optical wireless communication systems are examined. For indoor environment different configurations of transmitter and receiver positions are discussed. For *outdoor* environment, the factors affecting the link performance such as atmospheric turbulence are investigated in terms of channel modelling. Finally, both the indoor and *outdoor* channel models are respectively compared in terms of computation speed, complexity and accuracy of performance. The development of wireless communication systems operating in the region of infrared frequencies is driven by the demand of faster and denser data-rate transmission within a personal mobile communication environment. The characteristics and performance of the wireless optical channel are variables of the type of the link namely, directed or non-directed links and line of sight (LOS) or diffused links. The indoor optical wireless environment mainly benefits from non-directed diffused configurations as these links do not rely upon a line of sight path between the transmitter and the receiver eliminating the requirement of precise alignment. These links are robust against shadowing as the transmission is affected through high reflectivity of indoor objects such that a significant fraction of the transmitted signal arrives at the receiver after undergoing through a number of reflections. However, a large amount of transmitted optical power is required to provide a given signal coverage due to reflection loss. Additionally, the received signal is dispersed in time due to multipath propagation which results in reduced pulse amplitude, increased delay spread and causes intersymbol interference. Depending upon the size of indoor environment and the reflection coefficients of the objects within, multipath propagation of the transmitted signal puts a limitation on the maximum data-rate that can be transmitted. In the subsequent sections, we will examine different channel models that efficiently characterise the multipath propagation in a diffused indoor optical wireless communication link. In the *outdoor* optical wireless communication systems, the transceivers communicate directly through the air via point-to-point LOS links. In general, these links employ high-power lasers operating in class 3B band or above for maximum coverage at high data-rates. Due to the existence of direct LOS path between the transmitter and the receiver, the problems associated with multipath propagation are eliminated. As a result, very high datarates can be transmitted over a larger distance when compared with non-directed diffused systems. However, the factors affecting the performance of *outdoor* links are free space loss, clear air absorption, scattering and scintillation *noise* induced due to atmospheric turbulence:.

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Conference Paper

604 **Model calculations of wind noise measured in a flat surface under turbulent flow** — Yu J.,Raspet R.,Webster J., Dillion K. — 2008 Proceedings of the Noisecon ASME Ncad Ncad 2008 — 2009 — 0

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It has been proposed that microphones mounted flush on the ground surface should measure reduced wind *noise* levels since the microphone does not disturb the flow and the wind velocity at the ground is small. However, there are

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pressure fluctuations induced on the ground by the interaction of the turbulent fluctuations above the ground with the average wind velocity gradient. Model calculations of the pressure spectrum under different wind velocity profiles have been developed based on the theoretical work of Kraicnnan.1 Kraichnan calculated the turbulence-shear contribution to pressure fluctuations under boundary layer turbulence with an exponentially varying flow velocity profile. The calculations presented here incorporate measured *outdoor* spectra of the turbulence and different models of the velocity profile to investigate pressure fluctuation measurements outdoors at the ground surface. The predicted power spectral densities of the pressure are compared to each other and to measurements. The ultimate purpose of this research is to develop means to minimize wind *noise* interference in *acoustic* measurements. [Research supported by the U.S. Army TACOM-ARDEC at Picatinny Arsenal, NJ.] © 2008 ASME.

Article

☐ 605 **Effects of soil water potential and moisture content on sound speed** Lu Z.,Sabatier J.M. Soil Science Society of America Journal , 73(5) 2009 73

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To monitor and understand the temporal variations of *sound* speed due to the changes of soil physical properties such as soil water potential, moisture content, and soil temperature under natural conditions, a long-term field soil survey has been conducted. In this study, transmitter and receiver *acoustic* probes, consisting of ten (five pairs) and five transducers respectively, were inserted into a prepared trench containing a soil-sand mixture in an *outdoor* test site. Five time domain reflectometers, five tensiometers, and five *thermocouples* were buried during the trench-filling at the same depths as the *acoustic* transducers. Measurements of *sound* speed, soil temperature, soil moisture content, and water potential were performed continuously, along with the measurements of surface temperature and precipitation over a period of 2 yr. Analysis of the data shows that there is a power law relationship between the *sound* speed and water potential. It is also found that the water potential is the dominant influence on the *sound* speed whereas the moisture content and temperature have relatively minor impacts. Brutsaert and Luthins theory was employed to calculate the *sound* speed as a function of the effective stress. The theoretical predictions were compared with the experimental data and they are in good agreement. The study suggests that *sound* speed measurement could be used as a new and effective tool for water potential measurement. © Soil Science Society of America.

Conference Review

☐ 606 **International Conference on Fibrous Materials 2009** [No Authors Found] Proceedings of the Fiber Society 2009 Spring Conference 2009 0

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The proceedings contain 427 papers. The special focus in this conference is on Nanofibers, Technical Textiles, Biomedical Textiles, Textile Processing, Textile Composites, Textile Testing, Apparel Technology, Textile Marketing and Textile Management. The topics include: Filtration behavior of an original two-layered nonwoven geotextile; mechanical properties of novel stitched 3D hybrid basalt composite; the study on the air content of electrospun nanofiber nonwoven fabric; determine pore size distribution in nonwoven fabrics; measurement of the dynamic permeability of airbag fabrics; liquid absorption properties of super absorbent fiber based on acrylic copolymers; the ageing of polypropylene geotextiles under *outdoor* environment; scenario simulations of nonwoven webs and fabrics; research on shear performance of tackified woven fabrics; optimization of fibrous insulation; *thermal* bonding of polypropylene; improving phosphate glass fibre performance in composites; improved performance of bag filters through fabric surface modification; study of physical characteristics of industrial filter bag; morphology effects on effective properties of multiphase materials; application of an adaptive neuro-fuzzy system for predicting rotor spinning yarn quality properties; characteristics and technology optimizing of core-spun yarns on rotor spinning frame; effect of calcium salt treatment on the structure and properties of tussah silk and mulberry silk; new development of whisk broom fiber and its use in composite products; research on enzyme boiling out and bleaching technology based on flax roving; spinability of demestic new star fine denier meta-aramid fibre; spinning of basalt fiber filament/cotton composite yarns; study on the pretreatment process of kapok; study on the relation between quality and harvesting time of flax; technique and mechanism research on kenaf holoenzyme degumming; research

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on hydrolysis property of WSPET membrane; degradation of bamboo pulp in imidazole ionic liquid; effect of pretreating time and stretching speed to the slenderization process of yak hair fiber; effect of protease treatments on the prickle of a worsted woven wool blend fabric; electrospinning from the edge of an array of cascade rotating disks; mechanism analysis on compact yarn in compact spinning with pneumatic groove; numerical approach to the convergent point of rotor-spun composite yarn; optimization of pectinase production from fungus and its application in flax roving; research and development of jacquard fabric with double different faces; structure and properties of cotton yarn spun by false twist compact drafting method; study on additional twist of fiber band in compact district of compact spinning; effect on jute fibre by alkali and laccase treatment for textile use; electrostatic flocking of superfine fiber; preparation of macropores on the surface of rayon-based carbon fiber; research of a device for reducing hairiness on auto coner; plasma treatment with perfluoroheptane to impart liquid repellency to cotton fabrics; study on preparation and property of formaldehyde removing cotton fabric; research on laccase degumming of kenaf fiber; discussion of hydrolysis technology of the regenerated silk protein; optimization for the durability of crown needle; influence of electric field interference on double needles electrospinning; a new process for dissolution of cellulose in aqueous ionic liquids; multiple jets from a polymer bubble in electrospinning; optimized design on the structure parameters of a reducing hairiness nozzle; supervising and controlling system of spinning machines based on industrial Ethernet; optimizing producing conditions of chitin deacetylase; microprofile extrusion with a conformal fluidic layer and a wall slip condition; production of biocidal polypropylene fibers; carbon nanotube reinforced bamboo cellulose fibres; electrospun nanofibre yarns; wire coil needleless electrospinning of polymer nanofibers; modeling of multifilament polymer fiber melt-spinning; nanobubble dynamics and bubble electrospinning; identification of textile fiber with THz time-domain spectroscopy; structure and properties of milk protein/PVA blend fiber; study on composition and structure of soybean protein/viscose blend fiber; study on the tearing properties of PVC coated bi-axial warp knitted fabrics; friction testing between human skin and fabric; a method for measuring fabric emissivity; analysis of viscoelastic behavior of MVS cotton yarn; discussion and research of electronic testing system for raw silk; mathematical approximation of the stress-strain curves of fabrics; realization of the mathematic model for leno color effect; study on size locations of sized yarns using Keyence digital microscope; characterization for the compressing behavior of fiber assemblies; automatic detection of woven fabric defects by one-class support vector method; an measuring method of animal fibers diameter based on auto-focus technology; automatic identification of lyocell and cotton fibers; research on prediction of cotton yarn quality by artificial neural network; study on physicochemical properties and wearing characteristics of down; structure and *thermal* properties of cellulose acetate; low velocity impact behaviour of multilayer textile composites; evaluation of compression stocking transparency; a cotton cloth bioreactor with immobilized papain; comparison of PGA and PLA fibers; research on hydrolysis property of WSPET membrane; preparation and characterization of SnO2 nanofibers; surface analysis of the styrene-butyl acrylate copolymer latex film; novel technique to prepare porous and nano carbon fibers; study on the dispersion technics of the mixed nano-powders TiO2 and ZnO; study on the poly-lactic acid fiber treated by chitosan; new reusable template for synthesis of nano/micro polypyrrole tubes; crosslinking of electrospun fibrinogen nanofibers; the histocompatibility of polyurethane/superfine silk fibroin powder blend film; functional nanocoatings on textiles through layer-by-layer self assembly; energy-related applications of electrospun nanofibers; finite element simulation of the booster diaphragm; *thermal* insulation performance of triangle hollow fiber; structure and antibacterial analysis of silver fiber; study on the *acoustic* insulation of wool fiber assembly; study of the structure and properties of alginate fiber with antibacterial activity; application of activated carbon fibers in fuel cell cathode filter media; applications of silver sputtering technique in intimate apparel design; consumer behavior of apparel online-purchasing in shanghai and Seoul; study on brand strategy of small fashion enterprises; a study on the enterprise education in a fashion school; the development of *thermal* silk stockings and their potential market in China; quality supervision and technical regulation and standard of China domestic textiles; general situation of ancient textiles research in China; an adaptive MTM framework based on body classification; data designing of clothing pattern vector database; the research and construction of human apparel contour data plat; an analysis on the young female breast type; dynamic pressure comfort of three garment sleeve structures; research on apparel CIMS interface technology based on XML; study on sensory evaluation of color value on silk clothing; study on polarization trend of sports apparel design; natural anti-bacterial and health function underwear; fit and tight-fit garment design based on parameterized 3D sketching; research of costume merchandise layout based on AHP; application and properties of medical textiles in Chinese hospital; 3D textile scaffolds for blood vessel tissue engineering; performance evaluation of composite braided sutures; profiling of internal pressure exerted by medical bandages; eco-friendly pretreatments of jute and cotton blended fabric; application of composite dyeing of vegetable dyes on wool fabrics; study on the stripping of dyes on dyed cashmere; study on colore tone change of dyed cashmere fabric in finishing; study on

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extraction and purification process of capsicum red pigment; effect of garamite content on dyeable polypropylene (PP) fiber; the development of an oxidized corn starch size; application of computer aided design in digital printing; research on dyeing wool with two kinds of natural dyes; study on the stripping of dyes on dyed cashmere; dyeing properties of PVA/superfine wool powder blend filaments and colorfastness properties of naturally colored cotton grown in China to laundry.

Conference Paper

607 **Cooling system of outdoor cabinet using underground heat pipe** — Hong Y.,Ji S.,Zhai L.,Chen Q.,Bianco C. — INTELEC International Telecommunications Energy Conference Proceedings, 4664056 — 2008 — 3

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Currently the energy-saving strategy is one of the top priorities of Telecom Operators and Governments. In particular, the area of cooling has a heavy impact on the total electrical energy consumption, and needs to be optimized in order to increase overall performances, reducing both CAPEX/OPEX and the environmental impact. The present paper describes a new cooling solution for FTTC *outdoor* cabinets (where a significantly high amount of heat needs to be dissipated). This solution is based on geothermal cooling technique, that takes advantage of the ground heat (in this case "cold") source, which is always available and has an enormous potential. The geothermal cooling unit has been used in conjunction with forced air circulation within a sealed cabinet. The cabinet prototype is designed with an independent and integrated ventilation system which ensures that the *thermal* load generated within the cabinet itself is dissipated into soil and atmosphere. The aluminum-ammonia heat pipe technology is applied to this geotherm cooling unit. Experimental tests were carried out in real outfield. The cabinet was submitted to solar radiation and real ambient conditions. To carry out the analysis, a data acquisition system was set in the testing environment during a period of more than 3 months, over 40 measurement points were set to monitor temperatures, operating parameters and working loads. The experimental results show that, for a prototype cabinet with 700W heat load, this innovative solution cool the cabinet with COP value at about 110 for normal ambient condition of 30°C, which is comparatively higher than traditional air heat-exchanger solution (normally at 10-15). Moreover, the *acoustic noise* level is seriously reduced. The air flows through an independent underground ventilation system, and the air's temperature within the sealed cabinet can be reduced of about 5°C. The test data show that more than half of total consumption is dissipated into underground soil. Then, on a long-run perspective, the feasibility and the effectiveness of the described solution allow deep reduction of operational costs, with positive environmental impacts. ©2008 IEEE.

Conference Paper

608 **The variability of acoustical turbulence in the atmospheric boundary layer** — Cheinet S. — Proceedings European Conference on Noise Control — 2008 — 0

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Outdoor sound propagation is affected by small-scale turbulence, quantified through the refractive index structure parameter C_n^2. In this study, a Large Eddy Simulation (LES) is used to reveal the spatial and temporal variability of C_n^2 in the atmospheric convective boundary layer. A considerable variability is predicted, of almost two orders of magnitude at all heights. The regions of high C_n^2 match the convective plumes that drive the boundary layer dynamics. The impact of this variability on line-of-sight propagation and on *sound* scattering is illustrated. It is found to result in fluctuations of the considered *acoustical* diagnostics, which can in some cases induce major changes in the estimation of sensing performances.

Conference Paper

609 **Window usage, ventilation, and formaldehyde concentrations in New California homes: Summer field sessions** — Offermann F.J.,Robertson J.,Springer D.,Brennan S.,Woo T. — Iaq Conference — 2008 — 2

	Document title	Authors	Source	Year	Citations

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Concerns have been raised regarding whether homeowners use windows, doors, exhaust fans and other mechanical ventilation devices often enough to remove indoor air pollutants and excess moisture. In 2006-2007 we conducted a multi-season study of ventilation and IAQ in 108 new single-family, detached homes in California. This paper presents the ventilation and formaldehyde measurements from the Summer field sessions. A total of 10% of the 63 homes did not open their windows/doors at all during the 24-hour test period and 16% opened their windows/doors less than an average of 0.5 ft^2 (0.05 m^2). A total of 50% of the 62 homes with PFT measurements had *outdoor* air exchanges rates below 0.35 ach. A total 62% of the 61 homes with formaldehyde measurements had indoor concentrations exceeding the California ARB exposure guideline of 33 $\mu g/m^3$. We conclude that the new single-family detached homes in California are built relatively tight, and in those homes where the windows/doors are not opened for ventilation (e.g. for security, *noise*, odor, dust, *thermal* comfort concerns) the *outdoor* air exchange rates are typically low (e.g. 0.2 ach) and indoor concentrations of air contaminants with indoor sources such as formaldehyde can be significantly elevated. This study suggests that consideration should be given to installing mechanical *outdoor* air ventilations systems in new single-family residences to provide a dependable and continuous supply of *outdoor* air to the residence. © 2008 ASHRAE.

Conference Paper

610 **Terahertz target illumination fluctuation estimates derived from field measurements of atmospheric water vapor** — O'Brien S.G.,Tofsted D.H. — Proceedings of SPIE the International Society for Optical Engineering, 6949, 694906 — 2008 — 1

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Many terahertz imaging systems under development will be employed in *outdoor* environments, where spatial and temporal fluctuations of atmospheric absorbing species can affect image quality. Absorption across most of the terahertz band is dominated by water vapor. Active systems that illuminate targets with scanned ("flying spot") or floodlight terahertz sources will experience some spatial and temporal *noise* modulation of target plane irradiance due to pathintegrated inhomogeneities in the turbulent water vapor density field. We have analyzed data collected during field measurement campaigns conducted at the White Sands Missile Range during the spring and summer of 2007 for spectral characteristics and diurnal variations of water vapor fluctuations under dry to moderately humid synoptic conditions. The results of these analyses were then used to model the statistics of irradiance fluctuations that might be observed in the target plane of a THz imager under varying propagation conditions. The measurements acquired can also be compared with a statistical model of path-integrated absorptance considering either the evolution of the absorption with time or the effects of decorrelation in absorber for two angularly separated lines-of-sight.

Article • *Open access*

611 **Modeling growth of Matthiola incana in response to saline wastewaters differing in nitrogen level** — Grieve C.M.,Poss J.A., Shouse P.J.,Carter C.T. — Hortscience, 43(6) — 2008 — 11

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The capture and reuse of nutrient-rich greenhouse effluents may be an environmentally *sound* option for floriculture production, which would conserve fresh water resources and reduce off-site pollution of surface and groundwaters. This study was initiated in 24 *outdoor* lysimeters to determine effects of salinity and varying concentrations of nitrogen on the growth, yield, and ion relations of stock [Matthiola incana (L.) R. Br.] cultivar Cheerful White. The experiment was a 4 x 4 factorial, partially replicated design with four irrigation water salinities (2,5,8, and 11 $dS \cdot m^{-1}$) and four nitrate concentrations (2.5, 3.6, 5.4, and 7.1 $mmol \cdot L^{-1}$; N = 35, 50, 75, and 100 ppm). Ammonium nitrogen was included in the nutrient solutions. Stem lengths were measured three times weekly. Measurements at final harvest were stem and inflorescence lengths, stem and floret diameters, number of axillary buds and florets, and shoot and root fresh weights. Time course of stem elongation was quantified as a function of *thermal* time with a phasic growth model. Salinity significantly delayed initiation of the exponential growth phase, shortened its duration, and reduced the rate of plant development. The overall effect was to delay time to harvest of marketable stems. Although length

	Document title	Authors	Source	Year	Citations

of the flowering stems decreased with increasing salinity, marketable stems (≈60 cm) were produced in all treatments. Mineral ion relations in the plant tissues were influenced significantly, but independently, by both salinity and nitrogen. Leaf sodium, magnesium, and chlorine concentrations increased with increasing salinity; calcium and potassium decreased. In response to increasing external nitrogen, both potassium and chlorine decreased; sodium increased, whereas calcium and magnesium were unaffected. We conclude that in closed-loop irrigation systems, the nitrogen requirements for stock are low and that growers could minimize costs and limit off-site pollution by reducing nitrogen inputs.

Conference Paper

612 **Analyzing thermal environment efficiency of raised-floor Ondol with ventilation** — Sung J.-H.,Han S.-M., Kim Y.-M.,Sohn J.-Y. — Iaqvec 2007 Proceedings 6th International Conference on Indoor Air Quality Ventilation and Energy Conservation in Buildings Sustainable Built Environment, 1 — 2007 — 0

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A floor heating, so called 'Ondol', is widely used in korea, and it has been known that Ondol provids good *thermal* environment. [1] However, the impact *sound* through floor is great and ventilation rate depends on infiltration in rooms where the Ondol is applied.[2] It is reported that impact *sound* is reduced on raised-floor Ondol system.[3] This study aims at analyzing *thermal* environment efficiency of raised-floor Ondol with ventilation system which is an alternative plan about Ondol with insufficient ventilation. The study is performed at a residential house located on Yang-Pyoung, Kyoung-gi do in Korea. A size of sample house is 4.45m (L) x 3.35m (D) x 2.85 (H). Experimentation compared factors of *thermal* environment of each case; Ondol-heating only (Case A), introduction of *outdoor*-air (Case B), using raised-floor Ondol with ventilation system. Case A was the highest in every factor of *thermal* environment, and then case C, case B was. The case C is likely to effectively contribute to keep required indoor environment with better indoor air quality.

Conference Paper

613 **Numerical simulations of the natural ventilation design mix of elementary school classrooms** — Wu H.-H.,Wang F.-J., Chiang C.-M.,Lai C. — Ibpsa 2007 International Building Performance Simulation Association 2007 — 2007 — 0

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Most traditional elementary school classrooms here do not have air-conditioners and it often feel stuffy in classrooms located in the top floors. In addition, traditional classrooms widely have *noise* problem, so do the open educational classrooms. If the windows and doors are closed to reduce *noise*, it will result in the phenomenon of stuffiness, which largely prevents the external air entering the room. Therefore, it has become an urgent matter for building designers and educational administrators about how to solve the problem of ventilation as well as *thermal* comfort of elementary school classroom. CFD (Computational Fluid Dynamics) simulation method was conducted to investigate how the 6 natural ventilation design mix (Double Roof, CRSO (Covered Ridge with Sidewall Opening), roof ridge opening and window opening) influence indoor *thermal* environment at summer season and low *outdoor* wind-speed. Moreover, this research discussed how to use these designs to improve the natural ventilation efficiency of classroom. The results demonstrated that when collocating with Double Roof, roof ridge opening and CRSO, both ventilation efficiency and *thermal* comfort of the room are superior to other cases, no matter the room adapt staggered or conventional window layout. The staggered window openings can increase the buoyancy ventilation effect, compared

	Document title	Authors	Source	Year	Citations

with conventional window openings, they can increase the air change rate about 1.21 (1/h), equivalently increase by 1.5 times more.

Conference Paper

614 **Thermal-visible video fusion for moving target tracking and pedestrian classification** — Leykin A., Ran Y., Hammoud R. — Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 4270442 — 2007 — 68

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The paper presents a fusion-tracker and pedestrian classifier for color and *thermal* cameras. The tracker builds a background model as a multi-modal distribution of colors and temperatures. It is constructed as a particle filter that makes a number of informed reversible transformations to sample the model probability space in order to maximize posterior probability of the scene model. Observation likelihoods of moving objects account their 3D locations with respect to the camera and occlusions by other tracked objects as well as static obstacles. After capturing the coordinates and dimensions of moving objects we apply a pedestrian classifier based on periodic gait analysis. To separate humans from other moving objects, such as cars, we detect, in human gait, a symmetrical double helical pattern, that can then be analyzed using the Frieze Group theory. The results of tracking on color and *thermal* sequences demonstrate that our algorithm is robust to illumination *noise* and performs well in the *outdoor* environments. © 2007 IEEE.

Article

615 **Thermal comfort assessment in semi-outdoor environments: Application to comfort study in stadia** — Bouyer J., Vinet J., Delpech P., Carre S. — Journal of Wind Engineering and Industrial Aerodynamics, 95(9-11) — 2007 — 49

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The present study addresses *thermal* comfort assessment of *outdoor* and semi-*outdoor* environments. Two stadium case studies are used to demonstrate the potentiality of the approach that combines wind tunnel data and the calculation of human heat balance due to particular climatic environments. The *thermal* index PET (physiological equivalent temperature) is used to evaluate the *thermal* comfort in such complex environments. The specificities of stadium semi-*outdoor* spaces are summarised and the necessary assumptions made to apply the computation procedure are described. The approach includes assumptions on the *thermo*-physical phenomena as well as geometric computations. This work benefited from the development of an interactive design tool of built environments (*outdoor* urban surroundings): EVE (enriched virtual environments). It is a virtual reality platform developed at CSTB to help designers, architects and urban planners to evaluate the various options in competition regarding *acoustic*, visual, *thermal* and wind comfort of pedestrian in a particular urban environment. © 2007 Elsevier Ltd. All rights reserved.

Article

616 **Comparison of organic coating accelerated tests and natural weathering considering meteorological data** — Deflorian F., Rossi S., Fedrizzi L., Zanella C. — Progress in Organic Coatings, 59(3) — 2007 — 94

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The protective properties of organic coatings for *outdoor* applications are generally evaluated by means of accelerated laboratory tests, including electrochemical techniques. The coatings are stressed by different mechanical, chemical, *thermal* loads and the effects on the protective properties can be measured by using well established electrochemical techniques, like electrochemical impedance spectroscopy, electrochemical *noise*, etc. An open question is how these accelerated tests can be correlated with natural exposure in different environments. It is well

	Document title	Authors	Source	Year	Citations

known that many papers did not find any significant correlation between the natural exposure and different laboratory tests. This fact is due to many reasons, but one aspect very important is that the natural exposure is generally not sufficiently defined. The time of exposure (and the season) in a natural site is not sufficient to define the weathering, being the meteorological conditions variables considering different years. In this paper we proposed a new different approach. The starting point is a very simple consideration. We can try to define the natural weathering by considering a few different environmental parameters affecting the organic coatings properties: total amount of energy coming from UV radiation, time with relative humidity higher than a defined threshold (causing surface water condensation), environmental temperature higher that the polymer T_g, temperature range, etc. In this way, by carefully monitoring these parameters, it is interesting to verify if a correlation exists with some artificial accelerated tests (UV exposure, *thermal* cycling, salt spray chamber, etc.). This paper is a first attempt to apply this approach to a polyester coil coating for *outdoor* application. We considered different times of natural exposure in Trento (alpine region, north of Italy), monitoring at the same time all the meteorological parameters. The same coatings were tested in the laboratory using electrochemical impedance spectroscopy, in order to quantify the damage, after some different accelerated tests: *thermal* cycling, salt spray chamber and UV exposures. The results were compared with the impedance data measured after natural exposure and a correlation was investigated between the sum of the "natural stresses" and the sum of the "artificial stresses". The first results are very interesting and encouraging future work using the same approach on different materials and comparing several natural sites of exposure. © 2007 Elsevier B.V. All rights reserved.

Article

☐ 617 **Guidelines for improving natural daytime ventilation in an office building with a double-skin facade** — Gratia E.,De Herde A. — Solar Energy, 81(4) — 2007 — 81

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In recent years, there has been a great deal of interest in double-skin facades due to the advantages claimed for this technology in terms of energy saving in the cold season, protection from external *noise* and wind loads and their high-tech image. The advent of computers and other office equipment has increased the internal heat gains in most offices. Highly glazed facades, together with the extra heat gains from the electric lighting made necessary by deep floor plans and the wider use of false ceilings, have increased the risk of overheating. To preserve comfort and reduce cooling loads, it is important to apply natural cooling strategies, including natural ventilation. Some argue that double-skin facades are designed to improve natural ventilation in buildings by the stack effect, and to allow this even in situation in which it is generally not possible due to high *outdoor noise* levels and/or high wind speeds. But poor operation of the double-skin facade openings can generate disastrous scenarios such as the injection of hot air from the double-skin facade into the offices and the contamination of offices on the upper floors by used air from the offices on the lower floors. This article examines how natural ventilation can be utilised in an office building with a double-skin facade during a sunny summer's day. It mainly considers natural daytime ventilation in relation to the orientation of the double skin and the speed and direction of the wind. © 2006 Elsevier Ltd. All rights reserved.

Article

☐ 618 **Investigation and analysis on indoor environment of urban residential buildings in Chongqing** — Zheng J.,Shun J.,Zhou Y.-L., Yang C.-J. — Journal of Harbin Institute of Technology New Series, 14(SUPPL.) — 2007 — 0

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It is purposed to look into the actual conditions of the residential indoor environment in Chongqing during the autumn. Based on buildings' age and placement, twenty typically households were chosen and field measure was conducted in November 2005, these data were collected in living-room, kitchen, bedroom, toilet, etc. Finally, the characters of the indoor environment were obtained. Investigation revealed that, in this period, residential indoor *thermal* comfort level is better, but the *sound* environment is poor, with 42.68% of households dissatisfied with the *noise*. The indoor air is most polluted by formaldehyde seriously; and in the season *outdoor* concentration of certain

	Document title	Authors	Source	Year	Citations

pollutants (such as formaldehyde, etc.) are higher than indoor generally, therefore it is believed that residents should be strengthened ventilation to improve indoor air quality after that the *outdoor* air should be handled.

Article

619 **Temperature effect on tyre-road noise** — Anfosso-LedeE F., Pichaud Y. — Applied Acoustics, 68(1) — 2007 — 72

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Tyre-road *noise* emission decreases when the *outdoor* temperature increases, with a variation that can exceed -0.1 dB(A)/°C. This effect depends on tyre-road combination, but semi-generic corrections can improve the accuracy of tyre-road *noise* measurements. In this paper, the variation of pass-by *noise* level of a passenger car at 90 km/h with temperature is investigated, on seven types of road surfaces, under different temperature conditions. A good correlation between air, road surface and tyre temperature is outlined. A linear relationship between *noise* level and air temperature variations is observed for bituminous pavements, of about -0.1 dB(A)/°C, but reduced to -0.06 dB(A)/°C for pavements having porosity. No temperature effect is observed on cement concrete pavements. A spectral analysis shows that the temperature effect is highest in low and high frequency range, what can be explained by generating mechanisms rather than propagation. © 2006 Elsevier Ltd. All rights reserved.

Conference Paper

620 **Microphone noise when measuring in wind** — Van Den Berg G.P. — Euronoise 2006 the 6th European Conference on Noise Control Advanced Solutions for Noise Control — 2006 — 0

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Microphone *noise* induced by wind is the result of turbulence in the air. *Outdoor* air turbulence is created by friction and vertical temperature differences and can be quantified with a model containing expressions for surface roughness (friction) and atmospheric stability (*thermal*). The resulting pressure level as observed with a microphone can be calculated when the microphone is in wind over a relatively flat, constant roughness area. Microphone wind *noise* is important when doing measurements in relation to wind turbine *noise* and ambient background *sound*; both are measured at wind velocities up to 20 m/s. It can be shown that measurement results may be contaminated by wind-induced microphone *noise*. The level of wind-induced *noise* can be determined when the wind velocity is measured at two different heights.

Review

621 **Comparison of accelerated tests and natural weathering of protective organic coatings: Use of meteorological data | Confronto fra invecchiamento accelerato e naturale di rivestimenti organici protettivi: Uso di dati meteorologici** — Deflorian F.,Rossi S., Fedrizzi L.,Zanella C. — Metallurgia Italiana, 98(6) — 2006 — 0

Hide abstract Find It @ UCL View at Publisher Related documents

The protective properties of organic coatings for *outdoor* applications are generally evaluated by means of accelerated laboratory tests, including electrochemical techniques. The coatings are stressed by different mechanical, chemical, *thermal* loads and the effects on the protective properties can be measured by using well established electrochemical techniques, like electrochemical impedance spectroscopy, electrochemical *noise*, etc. An open question is how these accelerated tests can be correlated with natural exposure in different environments. It is well known that many papers did not found any significant correlation between the natural exposure and different laboratory tests. This fact is due to many reasons, but one aspect very important is that the natural exposure is generally not sufficiently defined. The time of exposure (and the season) in a natural site is not sufficient to define the weathering, being the meteorological conditions variables considering different years. In this paper we proposed a new different approach. The starting point is a very simple consideration. We can try to define the natural weathering

	Document title	Authors	Source	Year	Citations

by considering a few different environmental parameters affecting the organic coatings properties: total amount of energy coming from UV radiation, time with relative humidity higher than a defined threshold (causing surface water condensation), environmental temperature higher that the polymer Tg, temperature range, etc. In this way, by carefully monitoring these parameters, it is interesting to verify if a correlation exists with some artificial accelerated tests (UV exposure, *thermal* cycling, salt spray chamber, etc.). This paper is a first attempt to apply this approach to a polyester coil coating for *outdoor* application. We considered different times of natural exposure in Trento (alpine region, north of Italy), monitoring at the same time all the meteorological parameters (table 1). The same coatings were tested in the laboratory using Electrochemical Impedance Spectroscopy, in order to quantify the damage, after some different accelerated tests: *thermal* cycling (figures 6 and 7), salt spray chamber (figures 8 and 9) and UV exposures (figures 2 and 3). The results were compared with the impedance data measured after natural exposure (figures 2 and 3) and a correlation was investigated between the sum of the "natural stresses" and the sum of the "artificial stresses" The first results are very interesting and encouraging future work using the same approach on different materials and comparing several natural sites of exposure.

Article

☐ 622 **Wind-induced noise in a screened microphone** Van Den Berg G.P. Journal of the Acoustical Society of America , 119(2) 2006 63

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To reduce the effect of wind on a microphone, wind screens are used, usually made of foam or cloth. Although this measure to reduce wind *noise* is well known and widely used, it lacks a theoretical explanation. One possible explanation was the turbulent wake in the air flow behind the wind screen, another the pressure variations due to turbulence inherent in *outdoor* wind. In the present paper it is shown that atmospheric turbulence is indeed the cause of *outdoor* microphone wind *noise*, and the knowledge that is available in atmospheric boundary layer physics can be used to model the effects of atmospheric turbulence on a bare or screened microphone. A wind screen can be seen as a first order low pass filter for pressure variations due to atmospheric turbulence. Induced wind *noise* is thus determined by (average) wind speed and wind screen diameter, but also by factors related to frictional and *thermal* turbulence (roughness length and atmospheric stability). This approach leads to predicted spectral values of the induced "*sound*" pressure that match experimental data. The result can be used to obtain proper *outdoor* ambient wind related *sound* levels by correcting measured *sound* levels with calculated wind-induced pressure levels. © 2006 *Acoustical* Society of America.

Article

☐ 623 **Urban environment influence on natural ventilation potential** Ghiaus C.,Allard F., Santamouris M., Georgakis C.,Nicol F. Building and Environment , 41(4) 2006 114

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Natural ventilation is less effective in urban than in rural environment, especially in street canyons. Reduced wind velocity, urban heat island, *noise* and pollution, are considered to be important barriers to its application in urban environment. To quantify their effect on ventilation, wind, temperature, *noise* attenuation and *outdoor*-indoor pollution transfer were measured for a large variation range and for various types of urban configuration. The models obtained can be used in the initial stages of building design in order to assess the viability of natural ventilation in urban environment, especially in street canyons. © 2005 Elsevier Ltd. All rights reserved.

Conference Paper

	Document title	Authors	Source	Year	Citations
624	**Robust multi-pedestrian tracking in thermal-visible surveillance videos**	Leykin A.,Hammoud R.	Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition , 1640581	2006	45

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In this paper we introduce a system to track pedestrians using a combined input from RGB and *thermal* cameras. Two major contributions are presented here. First is the novel model of the scene background where each pixel is represented as a multi-modal distribution with the changing number of modalities for both color and *thermal* input. We demonstrate how to eliminate the influence of shadows with this type of fusion. Second, based on our background model we introduce a pedestrian tracker designed as a particle filter. We further develop a number of informed reversible transformations to sample the model probability space in order to maximize our model posterior probability. The novelty of our tracking approach also comes from a way we formulate observation likelihoods to account for 3D locations of the bodies with respect to the camera and occlusions by other tracked human bodies as well as static objects. The results of tracking on color and *thermal* sequences demonstrate that our algorithm is robust to illumination *noise* and performs well in the *outdoor* environments. © 2006 IEEE.

Article

	Document title	Authors	Source	Year	Citations
625	**Importance of being connected to the outside in the workplace**	Chau C.K.,Leung T.M., Law M.Y.	Architectural Science Review , 49(1)	2006	2

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This study examines the relative importance of environmental attributes and sensation of *outdoor* connection in the design of windows for a workplace by focusing on a sample of practicing building designers in Hong Kong. Discrete choice analysis using a combined visual image and textual and numerical figures was employed to reveal the importance being used by the designers in their evaluation of four performance attributes. Our analysis reveals that the designers placed equal importance on sensation of *outdoor* connection and *thermal* performance in window design for the workplace when the interiors of their buildings were open to a view surrounded by other office buildings. On the contrary, daylighting and *acoustic* performances were not considered to be important in their decision making process. This implies that the designers in Hong Kong had already started to translate their environmental awareness into appropriate design decisions by factoring in *thermal* energy saving as one of the important considerations for the window designs for the workplace. © 2006 Taylor & Francis Group, LLC.

Conference Paper

	Document title	Authors	Source	Year	Citations
626	**Thermal comfort assessment in semi-outdoor environments: Application to comfort study in stadia**	Bouyer J.,Vinet J., Delpech P.,Carre S.	Eacwe 2005 4th European and African Conference on Wind Engineering	2005	0

Document title	Authors	Source	Year	Citations

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The present study addresses *thermal* comfort assessment of *outdoor* and semi-*outdoor* environments. Two stadium case studies are used to demonstrate the potentiality of the approach that combines wind tunnel data and the calculation of human heat balance due to particular climatic environments. The *thermal* index PET (Physiological Equivalent Temperature) is used to evaluate the *thermal* comfort in such complex environments. The specificities of stadium semi-*outdoor* spaces are summarized and the necessary assumptions made to apply the computation procedure are described. The approach includes assumptions on the *thermo*-physical phenomena as well as geometric computations. This work benefited from the development of an interactive design tool of built environments (*outdoor* urban surroundings): EVE (Enriched Virtual Environments). EVE is a virtual reality platform developed at CSTB to help designers, architects and urban planners to evaluate the various options in competition regarding pedestrian *acoustic*, visual, *thermal* and wind comfort, of a particular urban environment. Copyright © 2005 ITAM.

Conference Paper

627 **Operation modes and economic performance study of 100KW microturbine building cooling, heating and power systems** — Huang J., Feng Z., Yue C., Liu L. — Proceedings of the ASME Turbo Expo, 5, GT2005-68277 — 2005 — 4

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Microturbines are one of the most promising DG (Distributed Generation) technologies for Building Cooling, Heating and Power (BCHP) systems. They have advantages over other kinds of heat engines in terms of atmospheric emissions, fuel flexibility, *noise*, size, and vibration levels. The characteristics of the overall microturbine BCHP system may be different under various system configurations and operation modes which depend on the consumers and *outdoor* conditions. In this paper, on the basis of the authors' previous work, the various possible operation modes are discussed under different external loads. The optimal operation modes are suggested to the microturbine BCHP schemes considering the various energy demands of a hotel in Xi'an. In the later part of this paper, the *thermodynamic* analyses and economic performance of microturbine BCHP system based on the first and the second laws of *thermodynamics* are carried out. The results show that even though the BCHP cogeneration system has the better economic performance both from the first and second laws evaluation, the benefits are affected by the operation modes because of the load uncertainty. The influence factors on the performance, such as power rate, natural gas price, operation time, are analyzed and compared for microturbine and gas engine BCHP systems. With the increase of natural gas price, or with the decrease of power rate, the economic performance advantages of both microturbine BCHP system and gas engine system are weakened, and the longer the cooling time, the shorter the payback periods for the system. Copyright © 2005 by ASME.

Conference Paper

628 **Illumination strategies to achieve effective indoor millimetre wave imaging for personnel screening applications** — Doyle R., Lyons B., Lettington A., ... McNaboe J., Curtin P. — Proceedings of SPIE the International Society for Optical Engineering, 5789, 14 — 2005 — 11

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The ability of millimetre-waves (mm-wave) to penetrate obscurants - be they clothing, fog etc. - enables unique imaging applications in areas such as security screening of personnel and landing aids for aircraft. When used in an *outdoor* application, the natural *thermal* contrast provided by cold sky reflections off of objects allow for direct imaging of a scene. Imaging at mm-wave frequencies in an indoor situation requires that a *thermal* contrast be generated in order to illuminate and detect objects of interest. In the case of a portal screening application the illumination needs to be provided over the imaged area in a uniform, omni-directional manner and at a sufficient level of contrast to achieve the desired signal to *noise* ratio at the sensor. The primary options are to generate this contrast by using active *noise* sources or to develop a passive *thermally* induced source of mm-wave energy. This paper describes the approaches taken to developing and implementing an indoor imaging configuration for a mm-wave camera that is to be used in people screening applications. The camera uses a patented mechanical scanning method to directly generate a raster frame image of portal dimensions. Imaging has been conducted at a range of

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frequencies with the main focus being on 94GHz operation. Experiences with both active and passive illumination schemes are described with conclusions on the merits or otherwise of each. The results of imaging trials demonstrate the potential for using mm-wave imaging in an indoor situation and example images illustrate the capability of the camera and the illumination methods when used for personnel screening.

Conference Paper

629 **Laboratory and field performance of megapixel QWIP focal plane arrays** — Goldberg A.,Choi K.K., Cho E.,McQuiston B. — Infrared Physics and Technology , 47(1-2) — 2005 — 10

Hide abstract Find It @ UCL View at Publisher Related documents

We present the results of laboratory and field testing of a very large format (1024 x 1024 pixels) focal plane array (FPA) made using quantum well infrared photodetector (QWIP) technology. The FPA was produced by QWIP Technologies, Inc. and tested at the Army Research Laboratory (ARL). Tests performed included spectral response, dark current, 3d-*noise*, minimum resolvable temperature (MRT) and optical crosstalk. The spectral response was found to peak at 8.55 μm. The device was found to have conversion efficiency close to 10% and the level of dark current was low enough to allow background limited infrared performance (BLIP) at 76 K. The pixel operability was in excess of 97% and there were no large cluster defects. The FPA was thinned to less than 20 μm to allow *thermal* cycling and sharp focus. We will show indoor and *outdoor* imagery acquired with the FPA. The FPA was installed in a camera system that was taken to a major field test. We will show initial results of the analysis of the image data collected at that test. Vehicles of military interest were imaged at ranges between 500 m and 5 km. Published by Elsevier B.V.

Conference Paper

630 **A coupling method for SPL predictions in complex situations using acoustic and micrometeorological numerical codes** — Gauvreau B.,Lihoreau B., Berengier M., Blanc-Benon P.,Calmet I. — Forum Acusticum Budapest 2005 4th European Congress on Acustics — 2005 — 2

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The modelling of *outdoor* long-range *sound* propagation implies to take into account complex propagation situations including mixed influence of ground characteristics (topography, obstacles, impedance, etc.) and atmospheric conditions (refraction and turbulence). We have developed a new PE code coupled with a micrometeorological code in order to calculate the *sound* propagation above a non flat ground for such realistic *outdoor* situations. Both codes are detailed in the present paper. Our *acoustic* code (MW-WAPE) can deal with different boundary conditions like the introduction of impedance jumps, thin screens or complex topography. Therefore, predicting long-range *sound* propagation over a non-urban site with complex propagation media requires the knowledge of micrometeorological fields in the lower part of the atmospheric boundary layer. Thus our PE code is coupled with a dedicated micrometeorological code (SUBMESO) which simulates wind and temperature fields over moderately complex terrain with high resolution. Its output data are used as input data for our PE code through a suited routine. This paper also presents some comparisons between numerical predictions and experimental data issued from the LCPC experimental (monitoring) site located at Saint-Berthevin (F). They show a very good agreement for both *acoustical* and micrometeorological predictions.

Conference Paper

631 **Minima controlled recursive averaging noise reduction for multi-aided inertial navigation of ground vehicles** — Liut B.,Adamst M., Ibanez-Guzman J. — 2005 IEEE Rsj International Conference on Intelligent Robots and Systems Iros, 1544966 — 2005 — 0

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Low-cost inertial measurement units (IMUs) are increasingly becoming commercially available and the use of IMUs in autonomous vehicle applications has increased rapidly in the past decade. IMUs are subject to various errors, such as biases, drifts, nonlinearities, scale factors and *noise*. The *noise* is produced by various sources, such as *thermal* and

	Document title	Authors	Source	Year	Citations

vibrational disturbances. *Noise* estimation is critical in accurate inertial navigation systems (INS). The main contribution of this paper is that a *noise* analysis of the raw accelerations measured by IMUs during signal presence (i.e, an acceleration caused by a specific force) or absence (when the IMU undergoes constant velocity) is carried out to reduce these *noise* components, which corrupt the inertial data. After *noise* reduction, multi-aiding information from odometry, a single-axis gyroscope and vehicle constraints is utilized to bound the error growth of the inertial data and produce a reliable *outdoor* localization system. Experimental results are presented to show the effectiveness of the *noise* reduction method and the improved accuracy of the multi-aided INS. © 2005 IEEE.

Article

632 **Occupancy implications of spatial, indoor-environmental, and organizational features of office spaces** — Mahdavi A.,Unzeitig U. — Building and Environment, 40(1) — 2005 — 39

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The quality of office spaces has been said to have a significant effect on the health, comfort, satisfaction, and productivity of office workers. Numerous studies and projects aim at documenting the postulated impact of office space quality on occupancy factors. Nonetheless, there is a need for further systematic research efforts and reliable empirical data in this area. This paper presents a study of a number of architectural offices in Vienna, Austria. Data pertaining to certain spatial and indoor-climatic features of these offices was collected via observation and snap-shot measurements. Simultaneously, an occupancy evaluation was performed via interviews with professionals who work in these offices. The observations and evaluations addressed the relationship between the workplace and the *outdoor* environment, the layout and refurbishment of the workplace, the technical infrastructure of the workplace, the communication patterns in the workplace, and the *thermal*, visual, and *acoustical* indoor climate. The "subjective" evaluations were compared with the "objective" measures of workplace architecture and indoor climate. © 2004 Elsevier Ltd. All rights reserved.

Conference Paper

633 **Adapative delay system (ADS) for sound reinforcement** — Smith E.A.,Detsky J.S. — Canadian Acoustics Acoustique Canadienne, 32(4) — 2004 — 0

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At concerts and presentations, the *sound* system must be carefully calibrated to ensure the entire audience can hear the presenters clearly. For both indoor and *outdoor* venues, this is done by positioning speakers throughout the audience to reinforce the *sound* produced on stage. This technique introduces an added complexity, whereby the electrical signal to the speakers in the crowd travels much faster than the *sound* wave coming from the stage. The Adaptive Delay System (ADS) for *Sound* Reinforcement is a new method for synchronizing the *sound* throughout the audience. Unlike existing methods, it does not require complex calculations when initially configuring the *sound* system. Furthermore, it is also capable of accounting for time-variant conditions such as wind, which are neglected in the current methods of speaker synchronization. Maximum length sequences, a special type of pseudo-random *noise*, are injected into a speaker for several seconds. A specially placed microphone picks up this *sound* which is then cross correlated with the original *noise* to determine the propagation delay. As this *sound* is barely audible it can be used during a concert to adaptively correct for changing conditions.

Conference Paper

634 **BEM-studies of sound propagation in viscous fluids** — Langer S. — Eccomas 2004 European Congress on Computational Methods in Applied Sciences and Engineering — 2004 — 2

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	Document title	Authors	Source	Year	Citations

Sound propagation in real media is involved with loss of energy. The loss of energy is caused by various damping mechanism, especially internal friction, *thermal* conduction, and molecular *acoustical* absorption. Each of these effects can be described by a decay coefficient ß. Considering this coefficient leads to a modified complex Helmholtz-equation (1), $p(x)_{,ii} + \bar{k}^2 p(x) = 0$ (1) describing the *sound* pressure p in a viscous fluid dependant on the frequency LO, speed of *sound* c, and the complex wave-number $\bar{k} = k + i\mu = \omega/c + i\mu$. Here, using the Boundary Element Method, a numerical study of the influences of viscosity on *outdoor sound* propagation is presented. With exemplary calculations the relevance of considering energy losses in air for *outdoor sound* propagation is discussed. Using the modified Helmholtz-equation a secondary effect appears: The uniqueness of the integral representation is positively influenced. This is confirmed in numerical studies.

Conference Paper

☐ 635 **Battery thermal management system** Cosley M.R.,Garcia M.P. INTELEC International Telecommunications Energy Conference Proceedings 2004 38

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Most VRLA batteries used in telecommunications applications are designed to operate in an indoor environment of 25°C. When placed in an *outdoor* environment, the batteries must be heated to prevent loss of capacity and must be ventilated to avoid build-up of hydrogen gas that occurs during charging that isn't properly controlled. The most difficult problem is maintaining the 25°C ambient temperature of the battery in an *outdoor* environment that can easily exceed 46°C. All current commercial solutions (*thermoelectric*, air conditioners, *thermosyphons*, PCM) require at least additional valuable site space to house the batteries as well as the expense and reliability issues associated with remote cabling, alarming and monitoring. We focused on the fact that the battery represents a large *thermal* mass and typically has different conductivities in all three directions due to the nature of their construction to develop a split cooling system that can keep the batteries at 25°C +/- 2°C via direct contact cooling. This avoids the inefficiency of air mover based systems that monitor and control the unit based off the airflow of the cooler itself. The issue of heating in cold ambient conditions, as well as, hydrogen outgassing is also addressed. The system can be used stand-alone or integrated into existing enclosures. Field performance data will be presented and discussed.

Conference Paper

☐ 636 **Right Hand Circularly Polarized (RHCP) effects on the measurements and solutions of the pseudolite-based indoor navigation systems** Jun H. Proceedings of the 17th International Technical Meeting of the Satellite Division of the Institute of Navigation Ion Gnss 2004 2004 0

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The GPS system uses Right-Hand Circularly Polarized signals. When the signal is received by a patch antenna, the measurement has a phase error term depending on the line of sight (LOS) vector from a transmitter to a receiver. The LOS vectors from GPS satellites to the reference or the users are almost same, and the RHCP effects are eliminated by double differencing method on the *outdoor* GPS. On the other hand, The LOS vectors from pseudolites to the reference and the user are different from each other and changed when the user moves, therefore, the RHCP effects are not removed even by double difference in the indoor navigation system. The RHCP effect is as big as several centimeters, which should not be negligible considering the error of the system is just several centimeter-level. This paper shows the RHCP effect on the asynchronous pseudolite-based indoor navigation system on simulations and experiments. The RHCP effect on time synchronization of the synchronous pseudolite-based indoor navigation system is also introduced. Then, it shows the RHCP-compensated results and the improvement on double difference measurements and position results.

Article

	Document title	Authors	Source	Year	Citations
637	**Neural network modelling of outdoor noise levels in a pilot area**	Avsar Y.,Saral A., Gonullu M.T., Arslankaya E.,Kurt U.	Turkish Journal of Engineering and Environmental Sciences, 28(3)	2004	18

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Noise measurements were made for 16 relevant *outdoor* points in the central campus area of Yildiz Technical University. All measurements were performed in accordance with Turkish Standards using an HD 9019 *sound* level meter. A new and sophisticated modelling technique called artificial neural networks was used to model the variation of *noise* levels from traffic around the campus area. The model inputs were the position of the measurement station, the geographical situation between the *noise* source and the measurement station, wind speed and direction, air temperature and relative humidity, and time of day. All parameters were measured throughout the study for 5 months. © Tubitak.

Conference Paper

	Document title	Authors	Source	Year	Citations
638	**Corrosion study of stainless steel bars in cracked concrete**	Yamaji T.,Hirasaki T., Takahashi R.,Mizuma S., Yamakawa M.	American Concrete Institute ACI Special Publication , SP-222	2004	0

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Corrosion of stainless steel bars in concrete was investigated using *sound* and pre-cracked concrete specimens. Three types of stainless steel were investigated, such as 18Cr, 18Cr-8Ni and 18Cr-12Ni-2.5Mo. Concrete specimens were exposed to two environments, where wetting and drying alternately repeated. One was in the *outdoor* with atmospheric temperature, and the other was in a controlled chamber, where the temperature was 60°C during wetting and 15°C during drying. The detail investigation was carried out after two years. No corrosion was observed on stainless steel bar in both *sound* and pre-cracked concrete exposed to *outdoor*. The maximum chloride ion concentration was 7.0 kg/m^3 for 18Cr-8Ni, 8.0 kg/m^3 for 18Cr-12Ni-2.5Mo and 6.0 kg/m^3 for 18Cr at the crack region of concrete. This result indicated that the chloride ion threshold level for stainless steel was larger than these values under marine environment with atmospheric temperature. No corrosion was observed on both 18Cr-12Ni-2.5Mo and 18Cr in both *sound* and pre-cracked concrete exposed to controlled chamber. However, corrosion was observed only for 18Cr-8Ni at the crack region, even when the chloride ion concentration at the crack region was 6.0 kg/m^3. This was considered to be due to the effect of a high temperature.

Article

	Document title	Authors	Source	Year	Citations
639	**Impact of portable air filtration units on exposure of haematology-oncology patients to airborne Aspergillus fumigatus spores under field conditions**	Engelhart S.,Hanfland J., Glasmacher A., ... Schmidt-Wolf I.G.H., Exner M.	Journal of Hospital Infection , 54(4)	2003	42

	Document title	Authors	Source	Year	Citations

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We undertook a one-year study to investigate the impact of the NSA model 7100A/B portable air filtration unit on exposure of haematology-oncology patients to airborne Aspergillus fumigatus spores under field conditions. Weekly measurements for airborne A. fumigatus were conducted in indoor and *outdoor* air, and surveillance for invasive aspergillosis was based on a combination of ward liaison, targeted chart review and consultation with the medical staff. The mean indoor A. fumigatus counts (8.1 cfu/m^3; range, <0.8 to 42 cfu/m^3) reflected the fungal load of *outdoor* air (9.4 cfu/m^3; range, <0.8 to 50 cfu/m^3), and were reduced by only about one third in rooms with portable air filtration units (5.3 cfu/m^3; range, <0.8 to 41 cfu/m^3). During the study period, a total of five cases (incidence density, 0.8 per 1000 patient-days) of invasive aspergillosis (one proven case, four suspected cases; case fatality rate 40%) were recorded. None of these five patients was allocated to a room with portable air filtration unit, however, the difference between incidence densities in rooms with and without portable air filtration units was non-significant (Fisher's exact test, P=0.33). Due to the *noise* level and *thermal* discomfort, patient compliance with the air filtration units was poor. We conclude that under field conditions this air filtration unit cannot be recommended for prevention of invasive aspergillosis in neutropenic haematology-oncology patients. © 2003 The Hospital Infection Society. Published by Elsevier Science Ltd. All rights reserved.

Article

640 **A note on the effect of gestation housing environment on approach test measures in gilts** — Marchant J.N., Bradshaw R.H., Marchant-Forde R.M., Broom D.M. — Applied Animal Behaviour Science, 80(4) — 2003 — 21

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Systems' welfare evaluation, including behavioural testing, is becoming increasingly popular in farm animal assurance schemes. The aims of this study were to investigate whether fairly short-term exposure to gestation housing systems, which varied in physical, environmental and human-input factors, influenced behavioural and physiological measures during a human approach test - often used to identify problems in human-animal interactions. Twenty-four Large White x Landrace gilts were initially subjected to identical human contact and daily husbandry. Forty-two days after service, the gilts were randomly assigned to either an indoor housing system (n = 16) or an *outdoor* housing system (n = 8), which differed physically and in the amount of human contact and daily husbandry. The indoor system used an electronic sow feeder (ESF), was more space-limited and *thermally*-controlled and had human contact centered on cleaning out. The *outdoor* system was more extensive, had much greater space accessible, was not *thermally*-controlled and had human contact that centered around feeding. The human approach test was carried out on all gilts 30-44 days after entry to the gestation system. At testing, each individual was fitted with heart rate monitor and then moved into a test arena. After 2 min an unfamiliar human entered the pen and stood motionless for 3 min against one wall and then approached the gilt and touched her snout. Throughout the experimental period, behaviour and *sound* within the test arena were recorded continuously. During the 2 min familiar-isation period, *outdoor* gilts had lower heart rates (108.2 bpm versus 123.7 bpm, $P < 0.05$) and tended to perform fewer short vocalisations (0.5 calls per min versus 3.4 calls per min, $P < 0.1$). *Outdoor*-housed gilts also carried out less locomotor behaviour (2.2 sections crossed versus 4.0 sections crossed, $P < 0.05$) and tended to perform fewer short (1.4 calls per min versus 5.0 calls per min, $P < 0.1$) and long vocalisations (0.2 calls per min versus 1.8 calls per min, $P < 0.1$) over the 3 min test period. *Outdoor* gilts tended to be slower to approach within 0.5 m of the human (69.9 s versus 19.3 s, $P < 0.1$) but they then took less extra time to make physical contact (3.3 s versus 52.7 s, $P < 0.1$). Mean heart rate was significantly lower in *outdoor* sows over the whole 3 min period (99.5 bpm versus 115.5 bpm, $P < 0.05$). The results demonstrate that short-term exposure to different housing systems did influence behavioural and physiological measures during a standard human approach test and thus, systems differences should be taken into account before making judgements about the human-animal relationship on any commercial farm, based on results of behavioural tests of this type. © 2002 Elsevier Science B.V. All rights reserved.

Article

	Document title	Authors	Source	Year	Citations
641	**A linearized Eulerian sound propagation model for studies of complex meteorological effects**	Blumrich R.,Heimann D.	Journal of the Acoustical Society of America , 112(2)	2002	110

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Outdoor sound propagation is significantly affected by the topography (including ground characteristics) and the state of the atmosphere. The atmosphere on its part is also influenced by the topography. A *sound* propagation model and a flow model based on a numerical integration of the linearized Euler equations have been developed to take these interactions into account. The output of the flow model enables the calculation of the *sound* propagation in a three-dimensionally inhomogeneous atmosphere. Rigid, partly reflective, or fully absorptive ground can be considered. The linearized Eulerian (LE) *sound* propagation model has been validated by means of four different scenarios. Calculations of *sound* fields above rigid and grass-covered ground including a homogeneous atmosphere deviate from analytic solutions by ≤1 dB in most parts of the computed domain. Calculations of *sound* propagation including wind and temperature gradients above rigid ground agree well with measured scale model data. Calculations of *sound* propagation over a screen including ground of finite impedance show little deviations to measured scale model data which are probably caused by an insufficient representation of the complex ground impedance. Further calculations included the effect of wind on shading by a screen. The results agree well with the measured scale model data. © 2002 *Acoustical* Society of America.

Conference Paper • *Open access*

	Document title	Authors	Source	Year	Citations
642	**A novel distance measurement system with a planar light wave circuit**	Harada M.,Doi M., Iwasaki Y.	Japanese Journal of Applied Physics Part 1 Regular Papers and Short Notes and Review Papers , 41(7 B)	2002	0

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We have developed a novel distance measurement (DM) system with a silica-based planar light wave circuit (PLC) and fiber optics for construction and civil engineering. The PLC contains *thermooptic* switches (TOSWs) that utilize the index change caused by the *thermooptic* effect. These switches are used for changing the light path and attenuating the light signal. The performance of the DM system was evaluated by the *outdoor* DM and linearity measurement. The *outdoor* DM at the distance of 104 m was achieved with the commercial retroreflective sheet and the measurement at the distance of 1 km with a cornercube prism is possible. The resolution of the system is about 1.7 mm (1σ). These results show our DM system fulfills the same required specifications as the conventional system that utilizes bulk optics. Moreover, the system has many advantages such as robustness, compactness, being *noise*-free, ease of alignment and stability.

Conference Paper

	Document title	Authors	Source	Year	Citations
643	**Overview of Integrating Dedicated Outdoor Air Systems with Parallel Terminal Systems**	Mumma S.A.	ASHRAE Winter Meeting Cd Technical and Symposium Papers	2001	1

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Integration of dedicated *outdoor* air (OA) systems with parallel terminal systems represents a huge shift in the standard engineering design paradigm. This paper will present some of the issues that guided the migration path toward separate dedicated OA ventilation systems and away from delivering the ventilation via single all-air variable air volume systems. The paper will then attempt to show that once a shift in the design paradigm in favor of a separate dedicated OA ventilation system has occurred, questions concerning the *thermodynamic* state of the delivered ventilation arise. Some of the boundary conditions that impact the best state of the delivered ventilation air will be addressed. Specifically, the first variable of interest is supply air temperature and its impact on the size and, hence, the cost of the terminal equipment and the terminal reheat energy. The second specific variable is the dew-point temperature (DPT) and the resultant latent cooling capability of the delivered ventilation air and its impact on

	Document title	Authors	Source	Year	Citations

the types of terminal equipment that may be practical. If in the paradigm shift the supply air DPT is dropped low enough to remove all of the space latent loads, then sensible-only terminal equipment, such as radiant ceiling panels, may become viable choices in some situations and eliminate the production of microbial problems in the spaces. Finally, this line of reasoning is extended further to note that when the space dew points are controlled with the dedicated OA, the terminal sensible equipment may use uninsulated sprinkler piping as the energy transport conduit. Companion papers (Mumma and Shank 2001; Shank and Mumma 2001; Janus 2001; Conroy and Mumma 2001) will address specific issues related to the overall integration of these concepts into a *sound* working design.

Article

644 **The attenuation of aircraft noise by wood stud walls** — Bradley J.S.,Birta J.A. — Canadian Acoustics Acoustique Canadienne , 29(3) — 2001 — 0

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Measurements were made according to the ASTM E90 procedure but with the frequency range of measurements extended from 50 to 5 kHz. The overall performance of the walls was rated using the ASTM *Outdoor*-Indoor Transmission Class (OITC). In general, it was found that the overall *sound* insulation of wood stud exterior walls is limited by poor performance at the low frequencies due to one of 2 types of low frequency resonance.

Conference Paper

645 **Overview of integrating dedicated outdoor air systems with parallel terminal systems** — Mumma S.A. — ASHRAE Transactions , 107 PART. 1 — 2001 — 60

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Integration of dedicated *outdoor* air (OA) systems with parallel terminal systems represents a huge shift in the standard engineering design paradigm. This paper will present some of the issues that guided the migration path toward separate dedicated OA ventilation systems and away from delivering the ventilation via single all-air variable air volume systems. The paper will then attempt to show that once a shift in the design paradigm in favor of a separate dedicated OA ventilation system has occurred, questions concerning the *thermodynamic* state of the delivered ventilation arise. Some of the boundary conditions that impact the best state of the delivered ventilation air will be addressed. Specifically, the first variable of interest is supply air temperature and its impact on the size and, hence, the cost of the terminal equipment and the terminal reheat energy. The second specific variable is the dew-point temperature (DPT) and the resultant latent cooling capability of the delivered ventilation air and its impact on the types of terminal equipment that may be practical. If in the paradigm shift the supply air DPT is dropped low enough to remove all of the space latent loads, then sensible-only terminal equipment, such as radiant ceiling panels, may become viable choices in some situations and eliminate the production of microbial problems in the spaces. Finally, this line of reasoning is extended further to note that when the space dew points are controlled with the dedicated OA, the terminal sensible equipment may use uninsulated sprinkler piping as the energy transport conduit. Companion papers (Mumma and Shank 2001; Shank and Mumma 2001; Janus 2001; Conroy and Mumma 2001) will address specific issues related to the overall integration of these concepts into a *sound* working design.

Article

646 **BLAST system capacity measurements at 2.44 GHz in suburban outdoor environment** — Gans M.J.,Amitay N., Yeh Y.S.,... Tran C., Adamiecki A. — IEEE Vehicular Technology Conference , 1(53ND) — 2001 — 14

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There are ever increasing demands, in wireless communications, for additional capacity to handle voice, data and wideband Internet applications. These demands are constrained by the bandwidth that was allocated to wireless communications. A novel unorthodox concept to substantially increase the spectral efficiency in highly scattering environments, named BLAST, was introduced in [1,2]. A measurement campaign was undertaken to assess the BLAST gains in spectral efficiency in the suburban *outdoor* environment for stationary subscribers. The measurements employed directive antennas to better control interference from adjacent cells. Extensive calibration methods,

	Document title	Authors	Source	Year	Citations

assisted by simulations, were developed to assure accurate results for the BLAST capacities of the measured remote subscriber sites.

Conference Paper

647 **Evaluation of load control strategies in an UTRA/FDD network**

Rave W., Kohler T., Voigt J., Fettweis G.

IEEE Vehicular Technology Conference , 4(53ND)

2001

20

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For a regular array of hexagonal cells in a suburban *outdoor* environment, we examined resource allocation schemes in the radio network controller (RNC) for the up- and downlink in a UMTS network. In the uplink the *noise* rise is the parameter of choice. In the downlink the power for different data rates and the connection as a whole serves as the means to control the network. Allocating this resource in the admission and congestion control, the throughput of a packet service with variable data rate was optimised.

Article • *Open access*

648 **Indoor environment quality in buildings and its impact on outdoor environment**

Roulet C.-A.

Energy and Buildings , 33(3)

2001

35

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The main purpose of buildings is to provide a comfortable living environment for their occupants. This includes, among others, *thermal*, visual and *acoustic* comfort as well as indoor air quality. Except during the 1950's and 1960's, it has always been considered important that an excess use of energy should be avoided in the construction and the management of a building, sometimes even at the cost of user comfort. Energy saving is, however, not the main purpose of the building. Indeed, if it were really so, the largest energy savings would be obtained by not erecting the building in the first place. Since the Rio conference, there have been more and more incentives to save energy and lower the impact of buildings on the environment. Therefore, there is no excuse for the building sector not to adopt a sustainable development policy. Some energy is required to control the indoor climate and indoor air quality. Therefore, it is often suspected that energy savings result in poorer indoor environment quality, or, on the contrary, that a high comfort level is the result of high technology and high energy consumption. This is not true. It is now generally admitted among building scientists that high quality energy services do not necessarily incur a high energy use, and that good environment quality can be obtained with a reasonable amount of energy and power, and with a low environmental impact. The presentation brings some evidence from past and current research to support this assertion.

Review

649 **IEA ANNEX 27: Evaluation and demonstration of domestic ventilation systems. Assessments on noise**

Op't Veld P., Passlack-Zwaans C.

Energy and Buildings , 27(3)

1998

7

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The acceptance and appreciation of ventilation systems is mainly determined by the perceived indoor air quality, *thermal* comfort and *noise*. *Noise* in relation to ventilation systems can be divided into three categories: (1) *outdoor noise* (entering the dwelling through ventilation openings, cracks, mechanical supply and exhaust openings, etc.); (2) *noise* generated by components of the ventilation system and: (3) the impact of ventilation systems on *sound* reduction of partitions (between dwellings, rooms, etc.). Depending on the type of ventilation system, one or more of these aspects are of concern. *Noise* related to the ventilation system and components, can result in turning off the ventilation system or closing vents, etc. This can have a negative influence on ventilation and indoor air quality. In the framework of IEA ANNEX 27, several *noise* aspects of domestic ventilation systems have heen evaluated. *Outdoor noise*: In *noise* loaded areas, the selection and the applicability of different types of ventilation systems are determined hy the *noise* level on the facades. A simplified tool is developed to select ventilation systems as a function of the required *noise* reduction of the facade, room dimensions and design and construction of the facade. System *noise*: Controlling *noise* levels caused by ventilation systems is in practice one of the most important factors to contribute to the satisfaction with a ventilation system. Air duct systems in dwellings transport *noise* generated by

	Document title	Authors	Source	Year	Citations

fans and aerodynamic *noise* generated by bends, control valves, grilles, etc. ANNEX 27 gives basic equations to estimate *sound* power levels of fans and grilles, indication of sixes for silencers and guidelines for design. Impact on *noise* reduction of partitions: The composite *sound* reduction is the result of different *sound* channels from one room to the other. One of this *sound* channels may be the ventilation system (cross-talk). Cross-talk can be brought about through the air duct system, overflow grilles and ventilation openings in partitions, duct transitions, etc. ANNEX 27 provides guidelines for *sound* reductions of partitions and insertion losses for ducts to eliminate the influence of cross-talk. @ 1998 Published by Elsevier Science S.A. All rights reserved.

Article

☐ 650 **Some practical considerations for predicting outdoor sound propagation in the presence of wind and temperature gradients** — Taherzadeh S.,Li K.M., Attenborough K. — Applied Acoustics, 54(1) — 1998 — 11

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The propagation of low-frequency *sound* above an impedance plane in a vertically stratified medium is considered. The combined effects due to the vector wind and temperature gradients on *sound* propagation are studied. Recently, a WKB-type approximation has been combined with turning point theory and applied to the efficient computation of the height-dependent Green's function, which, in turn, has been used in the fast field program (FFP). Although this method is applied to a temperature-stratified medium, it leads to a significant reduction in the computational time for predicting *sound* propagation outdoors. The present paper extends such a WKB-type approximation to include vector wind. Attention is focused on the development of efficient algorithms for numerical evaluation of the inverse Fourier transform and on considerations of numerical stability. The limitations of the approximation are examined for logarithmic profiles in particular. © 7997 Elsevier Science Ltd.

Article

☐ 651 **A study of energy use and satisfactory zone ventilation of different outdoor air ventilation strategies for terminal reheat variable air volume systems** — Reddy T.A.,Liu M., Claridge D.E. — Energy and Buildings , 29(1) — 1998 — 9

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Increased building indoor air quality (IAQ) complaints due to reduced *outdoor* air ventilation rates led to ASHRAE Standard 62-1989. Even though the stipulated standard total *outdoor* ventilation flow rate may be drawn into the HVAC system, *thermal* imbalances in the various zones of the building can lead to certain zones being starved of the specified ventilation flow rate, thereby creating localized IAQ problems. The objective of this paper is to compare the differences in energy use and ventilation air flow rates supplied to different zones in the building for three different practical *outdoor* air ventilation strategies all of which are identical in performance at design conditions but which differ under part-load operation. A simplified simulation methodology (which past studies have demonstrated to be useful for field evaluation of actual buildings) has been used to predict the heating and cooling energy use of a two-zone terminal reheat variable air volume (TRVAV) system during part-load operation specified by varying *outdoor* temperature and humidity conditions. The trade-off between *outdoor* air intake and energy use are studied for the following ventilation strategies for a typical 10,000 m^2 commercial building: (i) constant outside air intake based on a value 20% higher than the ASHRAE minimum ventilation rate, (ii) constant ventilation air intake fraction, and (iii) ventilation air intake based on the unfavorable zone requirements (even though the other zone may be over-ventilated). How this trade-off is impacted by building size has also been investigated. Finally, we use bin data for Dallas, TX (a moderately hot and humid location) and Seattle, WA (a mild location) in order to study the differences in energy use and zone ventilation flows of different ventilation strategies due to building location. The effect of economizer cycles and of varying ventilation strategies depending upon diurnal building schedules have not been considered in this study. The results of this study which are based on a simplified HVAC simulation approach are consistent with conclusions reached by other researchers Using more detailed simulation models. This suggests that *sound* and meaningful diagnostic insights of actual building performance and operating strategies can be obtained from such simplified simulations. © 1998 Elsevier Science S.A. All rights reserved.

Conference Paper

	Document title	Authors	Source	Year	Citations
652	**Large-scale survey of thermal comfort in office premises in Hong Kong**	Chan Daniel W.T., Burnett John, de Dear Richard J., Ng Stephen C.H.	ASHRAE Transactions, 104(Pt 1B)	1998	38

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Hong Kong is a densely populated city in which the service sector dominates. The significant *outdoor noise* pollution and subtropical climate severely restrict the opportunity for office premises to be naturally ventilated. The high energy consumption for space cooling and the demand for improved indoor *thermal* comfort conditions stimulated a large-scale survey of *thermal* comfort conditions in Hong Kong office premises. The neutral temperatures and preferred temperatures are found to be lower than those found in other studies in the tropics, with 60% of the surveyed subjects preferring a change of the *thermal* conditions in summer. The outcome provides for a better notion of *thermal* comfort, which can be imposed on design criteria. The results also add weight to the concern about the validity in the field of the traditional chamber test data presented by ASHRAE Standard 55-1992. It further suggests the potential for adopting an adaptive control algorithm for *thermal* comfort.

Article

653	**Silverstream eco-hydraulics flume: Hydraulic design and tests**	Nikora V.I.,Goring D.G., Biggs B.J.F.	New Zealand Journal of Marine and Freshwater Research, 32(4)	1998	29

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Hydraulic design details are presented of the Silverstream eco-hydraulics flume, which is a 12 m long by 75 cm wide tilting *outdoor* flume situated on the left bank of the Silverstream River, 30 km north of Christchurch, New Zealand. Results of testing, both mechanical and hydraulic, are presented and discussed. The mechanical adjustments required to accommodate the weight of water in the flume and *thermal* expansion and contraction of the frame (exacerbated by being outside) are described. Hydraulic calibration experiments using *Acoustic* Doppler Velocimetry indicate that there is a central portion at least 4 m long by 15–20 cm wide of the flume where the flow is uniform and 2-dimensional. Regions up stream and down stream of this area are affected by entrance and exit conditions. Close to the side walls, cells of helical secondary currents were found, and these appear to be separated from the zone of 2-dimensional flow by narrow regions with anomalous turbulence properties. The “frozen” turbulence hypothesis was proven true by the calibration experiments. These findings can have implications in eco-hydraulic experiments on organism-flow interactions, mass transfer, and uptake processes. © 1998, Copyright Taylor & Francis Group, LLC.

Conference Paper

654	**Effects of refraction on short range outdoor propagation over asphalt surfaces**	Hartwig Troy, Bolton J.Stuart	Proceedings National Conference on Noise Control Engineering, 1	1997	0

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The effects of wind and temperature gradients on short distance propagation measurements over asphalt surfaces were studied in the context of two-microphone method. In this paper, the fundamental equations governing *outdoor sound* propagation are presented, together with the methodology that was used to model refraction. The experimental verification of the methodology are also presented.

Article

655	**Long-term testing of a low-loss 800 kVPPLP-insulated SCFF cable**	Couderc D.	IEEE Power Engineering Review, 16(1)	1996	0

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In 1986, the government of Québec asked Hydro-Québec to study the feasibility of under-river passages for the St. Lawrence river crossings of future EHV lines. To meet this requirement, the utility launched a program to identify and

Document title	Authors	Source	Year	Citations

validate the most promising technologies for the 800-kV ac cable class. The development of an 800-kV PPLP-insulated fluid-filled cable and its accessories started in 1990 in collaboration with Sumitomo Electric Industries, Ltd.. Under this development program, the cable and its accessories were subjected to a series of tests at IREQ's high voltage laboratory and cable test facility. A type test of approximately two months was completed successfully at the end of 1991, following which a long-term test was initiated and completed one year later. This pa per describes the main developments of the long-term test program and reports the results obtained. A 9-segment Milliken type conductor having a cross section of 2000 mm was selected for the required rated current. The PPLP insulation (30 mm thick) is graded with a 1-mm layer of thin high-density kraft paper on the conductor screen to mitigate the maximum electrical stress. The test line for the long-term test is approximately 130 m long and includes two *outdoor* sealing ends, two SF6 switchgear sealing ends, and one straight joint with an insulated flange. The hairpin configuration of the cable run was chosen for its compactness, and especially for its suitability to simulate, in the straight section, the operating conditions of a two-cable phase m a typical under-river tunnel crossing. The temperatures measured by 20 *thermocouples* installed at precise locations in the conductor oil channel were relayed by an infrared transmitter to the control room. The test temperatures and voltages chosen to simulate approximately 30 years of service with a 40-week long test were based on a well known approach. The necessity to take into account the accessory lifetime and the complex effects of *thermal* cycling on the insulation of both the cable and the accessories suggested a conservative approach in favor of a value of 12 for the voltage exponent. The conductor temperature of the cable and the joint is kept at 90°C and 100°C as the test voltage is raised in three steps from 1.0 Um to 1.3 Urn. Each cycle of the cycling periods lasts 7 days. The temperature is raised to the maximum conductor temperature (90°C or 100°C) in approximately 8 h and kept at this value for 5.5 days. Follows approximately one day of forced cooling to bring the conductor temperature down by 40°C to 50°C. The loss tangent of the test line was measured monthly and the values obtained were within design specifications (percent). The gas content in the oil sampled at the five accessories indicates that the insulation aging was essentially *thermal* in origin andthat no abnormal behavior such as partial discharge activity was present. The test line was submitted to lightning impulse and switching surge tests following the long-term test. The conductor temperature of the cable and the joint was kept at 100°C during these tests. The cable and the accessories wisthstood successfully impulse voltages of ±2160 kV (BIL+20 percent) and switching surge voltages of-2170 kV. Since no failure occurred in the test line, dissections of the accessories and 19 sections of the cable were done with great care in order to inspect the insulation for any signs of defect or incipient problems. The results obtained revealed the insulation in both the cable and the accessories was *sound*. No signs of core movement at the joint could be observed, and the registration of tapes in bends and various mechanically stressed areas were within factory tolerances. In conclusion, the test program, including the type tests, showed very good performance of the PPLP-insulated cable and accessories designed to satisfy service conditions and insulation levels required for future 765-kV St. Lawrence under-river crossings. The results yielded by the long-term test confirm the conclusion reached after the type tests, namely that no further requirements or design modifications would be necessary.

Article

☐ 656 **Application of sound-absorbent plastic to acoustic paneling** — Ootsuta Katsuhisa, Tani Shuichi — Mitsubishi Electric Advance, 74 — 1996 — 5

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A durable high-performance *acoustic* panel was developed using a special *sound*-absorbent plastic *acoustic*-insulation material formed by subjecting *thermoplastic* resin beads to heat and pressure. The plastic is incorporated into a module that enhances its *sound*-deadening properties. Rugged, weather-resistant design and excellent longevity make the paneling suited to both indoor and *outdoor* use.

Article

☐ 657 **Condensation process in external concrete walls under random fluctuation of outdoor Temperature—Fundamental study on heat and moisture behavior by Fokker—Planck equation** — Hokoi S.,Matsumoto M. — HVAC and R Research, 2(4) — 1996 — 3

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To investigate the condensation process in a wall composed of porous materials under randomly changing *outdoor* temperatures, a method is proposed to calculate the probability density of moisture contents. By modeling *outdoor*

	Document title	Authors	Source	Year	Citations

temperature as a filtered Gaussian white *noise*, the heat and moisture transfer processes, and the stochastic process of *outdoor* temperature can be expressed by state space equations. The Fokker-Planck equation is derived by making use of this state space expression. A wall made of autoclaved, low density concrete is analysed, and the probability density of the moisture content is obtained. This simple example clarifies the influence of random *outdoor* temperature on the condensation process. © 1996 Taylor & Francis Group, LLC.

Article

658 **The use of electrochemical noise methods (ENM) to study thick, high impedance coatings** — Bierwagen G.P., Jeffcoate C.S., Li J., ... Tallman D.E., Mills D.J. — Progress in Organic Coatings, 29(1-4) — 1996 — 49

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Thick, high impedance organic coatings are those class of coatings used to provide corrosion protection to naval vessels, pipelines, gasoline storage tanks, and other large structures such as bridges and plant structures. These coatings, especially the newest generations now being used in practice, can provide exceptional protection and lifetime of performance such that properly and accurately assessing and differentiating among competing coatings is a very difficult task. The standard protocol of salt fog testing (ASTM B117), immersion testing, and *outdoor* exposure in a corrosive environment with subjective evaluation of a coating's performance during and after testing, does not adequately rank and predict coating lifetimes for new coating systems, especially for the environmentally compliant coating systems such as powder coatings (especially the thick, fusion bonded epoxy (FBE) coatings used for pipelines), two component epoxy and urethane coatings and waterborne coatings. New, objective test methods are desperately needed by users and manufacturers of coatings. A relatively new electrochemical test procedure, electrochemical *noise* methods (ENM), as developed by Skerry and Eden, has been shown in our laboratory to be very successful in the ranking and prediction of relative coating performance. We have used the method successfully on naval ship coatings, several pipeline coatings and other related systems, and Skerry has used them successfully on industrial maintenance coatings. We have used these methods in conjunction with electrochemical impedance spectroscopy, d.c. resistance measurements and cyclic salt fog testing of the Prohesion™ type. In our studies of pipeline coatings, we needed to investigate *thermal* effects because of their extended range of use temperature. In these studies, we have discovered that electrochemical methods can be used for an in situ measurement of the T_g of coatings in electrolyte immersion. Further, the 'plasticizing' effect of aqueous electrolyte absorption as well as its relative irreversibility has been shown. For all coatings studied, ENM provided useful, objective, numerical data which rapidly ranks coatings and provides useful information on the relative lifetime prediction of coatings which may provide up to 30 years of service.

Article

659 **Continuous Automated Measurement of Gaseous Nitrous and Nitric Acids and Particulate Nitrite and Nitrate** — Simon P.K., Dasgupta P.K. — Environmental Science and Technology, 29(6) — 1995 — 113

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An automated measurement system for HONO(g) and HNO_3(g) along with particulate nitrite and nitrate is described. The system alternately measures the gas phase and aerosol phase constituents. The gases are collected by a parallel plate wet denuder, and the effluent aerosol is collected by steam condensation, followed by impaction/*thermophoresis*. The fully automated instrument provides separate but quantitative collection of gases and particles at relatively high flow rates and displays very low interference from NO and NO_2. Coupled to an anion chromatograph, the detection limit estimated for a signal equivalent to that of three times the *noise* level of the blank is 110 parts per quadrillion (ppq, 10^{-15} atm) and 230 ppq of gaseous HONO and HNO_3, respectively, and 230 and 650 pg/m^3 aerosol nitrite and nitrate, respectively, for a 8-min integrated sample. The instrument operates over prolonged periods with minimal attention. Data are reported for *outdoor* ambient air measurements and for indoor air with kerosene-fueled space heaters in operation. © 1995, American Chemical Society. All rights reserved.

Article

	Document title	Authors	Source	Year	Citations
660	**Effect of thermal turbulence on sound propagation in the atmosphere in the presence of a shadow zone | Influence d'une turbulence thermique sur la propagation du son dans l'atmosphere en presence d'une zone d'ombre**	Chevret P., Blanc-Benon Ph.	Journal De Physique, 4(5 pt 2)	1994	0

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A numerical technique for simulating the *acoustic* wave propagation through a turbulent atmosphere is introduced. We first generate a random homogeneous and isotropic medium by creating N independent realizations in terms of discrete Fourier modes. Then, for each realization, we solve a parabolic equation using a Split step Pade algorithm. Finally, we calculate the mean intensity of the *acoustic* field and its phase and amplitude fluctuations. Our numerical results are compared with *outdoor* measurements.

Article

	Document title	Authors	Source	Year	Citations
661	**Research and education in physical acoustics at the University of Mississippi, USA**	Bass H.E.	Applied Acoustics, 41(3)	1994	3

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The University of Mississippi provides graduate education for a significant fraction of US students pursuing careers in physical *acoustics*. The education program emphasizes traditional physics coursework with research opportunities in *acoustics*. Research is conducted in a new, dedicated state-of-the-art research facility with access to modern data acquisition and analysis systems. Research projects include agroacoustics, *outdoor sound* propagation, *thermoacoustics*, *acoustically* active surfaces, active *noise* control, nondestructive testing and machine perception and non-linear *acoustics*. © 1994.

Conference Paper

	Document title	Authors	Source	Year	Citations
662	**Thermal and acoustical performance of `buffer rooms'**	Mahdavi A.	ASHRAE Transactions, 99(pt 1)	1993	4

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The term `buffer room' refers in this context to spaces built between *thermally*, visually, and *acoustically* `controlled' indoor rooms and the `noncontrollable' *outdoor* environment. Examples of buffer rooms are sunrooms, atria, (enclosed) staircases, and air locks. In a long-term research effort carried out in Austria, buffer rooms were studied with regard to their hygrothermal and *acoustical* performance within a human-ecological framework. Special attention was paid to the problems of temperature fluctuations and risk of overheating, ventilation rates, and humidity control as well as *sound* transmission. The research agenda included studies under controlled conditions in SHA, a facility dedicated to building physics research in Vienna, Austria, as well as field investigations. This paper gives a summarized overview of the content and results of some of these studies, focusing on the issues of *thermal* performance as well as the *acoustical* insulation effect of sunrooms and its relationship to natural ventilation.

Conference Paper

	Document title	Authors	Source	Year	Citations
663	**Variation of 222Rn concentration in outdoor air due to variation of the atmospheric boundary layer**	Kataoka T., Tsukamoto O., Yunoki E., ... Sahashi K., Fujii S.	Radiation Protection Dosimetry, 45(1-4 SUPPL.)	1992	11

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Variation of ^{222}Rn concentration in *outdoor* air, returns of a monostatic *acoustic* sounder and sensible heat flux are simultaneously observed at Okayama-shi in Japan. The sensible heat flux is measured with an eddy correlation method using a sonic anemometer and a copper-constantan *thermocouple thermometer*. Using height of a surface-based inversion layer obtained by the *acoustic* sounder returns and two ^{222}Rn concentrations during the presence of the surface-based inversion layer, the exhalation rate of ^{222}Rn at this site is estimated to be about 0.01 $Bq.m^{-2}.s^{-1}$, which agrees well with the exhalation rate of ^{222}Rn estimated at Nagoya-shi in Japan. Using the estimated exhalation rates of ^{222}Rn and the variation of the ^{222}Rn concentration in the daytime, heights of mixing layers are estimated. These heights of the mixing layers are larger than the height of the mixing layer estimated using the sensible heat flux and the representative vertical profile of the autumnal night time air temperature in this area.

	Document title	Authors	Source	Year	Citations
Article					
664	**Environmental Conditions in a Recently Constructed Office Building Before and After the Implementation of Energy Conservation Measures**	Farant J.-P.,Baldwin M., De Repentigny F.,Robb R.	Applied Occupational and Environmental Hygiene, 7(2)	1992	3

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This study documents environmental changes in a new office complex during the period of initial occupancy and before and after the implementation of energy conservation measures. The parameters measured as part of a comprehensive monitoring program were ambient air temperature, humidity, carbon dioxide, carbon monoxide, nitrogen dioxide, formaldehyde, total volatile organic chemicals (TVOCs), airborne particulates, *noise*, illumination, air velocity, and *outdoor* air supply. Three separate surveys were conducted during the first 18 months of occupancy. Energy conservation measures were introduced prior to the third survey. The changes in *thermal* conditions, the increase in carbon dioxide and TVOC concentrations, and the rise in occupant complaints during this period all attested to a deterioration of the indoor environment due to the ventilation systems' inability to accommodate the building's increased population and the implementation of energy conservation measures. Facility design engineers and building operators should use a similar approach, based on the monitoring of key environmental parameters, to identify and correct design shortcomings that can cause problems during the critical phases of a building's existence. This would ensure that occupant dissatisfaction with his/her work environment would not become entrenched. The study results also indicated that TVOCs originating from the outgassing of construction materials and furnishings effectively stabilized after one year of continuous operation of the ventilation systems and that occupant activities were important sources of these contaminants. © 1992 Taylor & Francis Group, LLC.

Conference Paper					
665	**On the asymptotic properties of a residential heater/air conditioner duty cycle**	Noureddine A.H., Alouani A.T.	Conference Proceedings IEEE SOUTHEASTCON, 1	1991	0

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The asymptotic properties of two probabilistic solutions for the duty cycle of a residential *thermostat* controlled heater/air conditioner unit are considered. The probabilistic solutions were derived previously from a physical *thermodynamical* model of the indoor temperature of a residential house as seen by a *thermostat*. The duty cycle solutions considered are functions of the house parameters, *outdoor* temperature, and the random effects of household activities (driving *noise* process). It is shown that the conventional (deterministic) duty cycle represents, as expected, a special case of the two solutions.

Article					
666	**An Atmospheric Optical Ring Network**	Binder B.T.,Yu P.T., Shapiro J.H.,Bounds J.K.	IEEE Transactions on Communications, 38(1)	1990	13

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An experimental 10 Mbits/s token-ring local area network (LAN) that uses atmospheric optical communication links is described. Objective and subjective performance results are reported for operation over a 170 m-long *outdoor* path. These data support the following hypotheses. Atmospheric optical communications may find LAN applications as bridges between buildings containing cable subnetworks or as temporary quick-connects for new outlying hosts for which cable runs are initially unavailable. LAN protocols for networks incorporating atmospheric optical communications should be modified to best insulate users from the occasional error bursts and bad-weather outages suffered by such links. © 1990 IEEE

Article					

	Document title	Authors	Source	Year	Citations
667	**Influence of environmental conditions on acoustical properties of sea ice**	Jezek K.C.	Journal of the Acoustical Society of America , 88(4)	1990	16

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Sonar echo amplitude data have been collected at carrier frequencies of 188 and 120 kHz from the underside of different sea ice types. Histograms of normal incidence echo amplitudes were formed from over 90 samples of each ice type. Experiments were conducted on saline ice grown in an *outdoor* pond under relatively controlled conditions at the USA Cold Regions Research and Engineering Laboratory (CRREL) and on the sea ice cover in the Fram Strait. Analysis shows marked variations (about a factor of 5) in the magnitude of the coherent reflection coefficients as congelation ice at the bottom of an ice sheet evolves from a growing dendritic interface to an ablating, *thermally* altered interface. Larger differences (about a factor of 10) are observed between growing congelation ice and slush ice, used to simulate frazil. These results indicate that important variations in *acoustic* regime exist in areas where different ice types are intermingled. © 1990, *Acoustical* Society of America. All rights reserved.

Conference Paper

	Document title	Authors	Source	Year	Citations
668	**Indoor Air Quality Diagnostics Qualitative and Quantitative Procedures to Improve Environmental Conditions**	Woods J.E.,Morey P.R., Rask D.R.	ASTM Special Technical Publication , STP 1002	1989	2

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Procedures conventionally used in industrial hygiene are often inappropriate for application in non-industrial environments. More sensmve procedures are needed, since comfort, occupant well-being, and general population susceptibilities are parameters not now addressed by occupational threshold limit values (TLVs). A common result is that compliance with industrial hygiene standards may be demonstrated, but complaints from occupants persist. A new discipline, building diagnostics, is useful in addressing these more sensitive parameters. An application of this discipline to the evaluation of indoor air quality is presented in this paper. A fundamental aspect to building diagnostics is the need to understand the relationships that exist between the building, its systems, and its occupants. This understanding is developed initially through qualitative diagnostics in which the scope of investigation is defined, appropriate environmental performance criteria are determined, the nature of any existing complaints is reviewed, and an engineering assessment of the HVAC system is conducted. From these qualitative data, a set of hypotheses is developed in terms of the inadequacies of the performance or the opportunities for improvement. Based on these hypotheses, system performance under various loads is predicted and ways to mitigate and improve performance are recommended. An important advantage of this approach is that many problems can be solved at this stage before measured data are required. However, in some cases, objective and subjective measurements may be reqmred to confirm hypotheses and to substantiate or modify the previous recommendauons. In these cases, quantitative diagnostics is indicated. Because of the cost involved in this phase, measurement locations must be identified carefully. We find that five locations in a functional area of the building are sufficient to characterize system performance and occupant response: locations of least ventilation effectiveness, strongest contaminant source, most susceptible occupants, a "control location," and an *outdoor* location. Only those quantitatwe measurements appropriate to the hypotheses are taken; these may include air quality, *thermal*, lighting, and *acoustics* parameters. Simultaneously, questionnaires are also used to measure human responses to the environmental conditions within the occupied spaces. These quantitative data are analyzed for compliance wtth the performance criteria. Recommendations for modifications are then made which address employee concerns, architectural concerns, and cost-effective operation.

Conference Paper

	Document title	Authors	Source	Year	Citations
669	**PUBLISHED RESULTS OF THE ARI CERTIFIED SOUND RATING PROGRAM.**	Potter Allen	Proceedings National Conference on Noise Control Engineering	1983	0

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The Air-Conditioning and Refrigeration Institute (ARI) has completed over a dozen years of publishing certified *sound* ratings for *outdoor* unitary equipment. This program covers the air-cooled *outdoor* portions of factory-made

	Document title	Authors	Source	Year	Citations

mechanical air-conditioning equipment or heat pumps rated in cooling capacity at less than 135,000 British *thermal* units per hour (Btuh). Beginning in 1983, ARI *sound* ratings for this equipment are in terms of a different descriptor than that previously used consistently from 1971 through 1982. The empirical relation between the two ratings, and also the inevitable correlation between *sound* output and *thermal* capacity is examined. It is shown that the *sound* of the *outdoor* portion of air-conditioning equipment varies substantially due to factors other than *thermal* rating. (For the same *thermal* capacity, *sound* ratings are distributed with a standard deviation on the order of 3 decibels.) Within such variation due to other factors, exists a strong positive correlation between *sound* output and *thermal* capacity. A relationship of the form L equals A plus B log(Q) is applicable.

Conference Paper

☐ 670 **NOISE CONTROL IN A VERY LARGE HALL OF A SHIPYARD.** Schuller Willem M., Straatsma Hendrik | Proceedings International Conference on Noise Control Engineering, 2 | 1983 | 0

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Article

☐ 671 **PROJECT T-I-D-E, TEMPERATURE INVERSION DETECTION EQUIPMENT.** de Graffenried Albert L. | IEEE Intercon Tech Pap 973 | 1973 | 0

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To detect the birth, growth rate, and break-up of ground-based atmospheric temperature-inversion layers (so important in ground-level airpollution) a 4-foot parabolic reflector with a 4 multiplied by 4 rectangular array of microphones at its image plane was used as the receiver on a 0. 6-mile *outdoor* sonic range. *Sound*-source image motion, as well as amplitude and phase modulation of 3. 4 khz signals were detected and interpreted. The mathematics and physics of the growth of the ground-based atmospheric temperature inversion layer are examined, and system errors are presented. Some reasons for initiating this project are listed.

Article

☐ 672 **Sheets made of Makrolon, (Platten aus dem Chemiewerkstoff Makrolon)** BEICHER H | Plastverarbeiter, 22(6) | 1971 | 0

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General remarks related to development and chemistry of West Germany produced polycarbonate, are followed by description of commercially available polycarbonate sheets, which are used for the production of high- grade plastics components and engineering moldings with close tolerance. The first part of this report deals primarily with performance problems- mechanical properties, *thermal* behavior, flame- retardant behavior, heat transfer, light transmittance, *outdoor* resistance, electrical and *acoustic* properties, dye- repellency, toxicologic properties, resistance to micro- organisms and humic acids, water vapor and gas permeability, behavior under high- energy radiation.

Article

☐ 673 **Radio wave scattering from the ionosphere** Farley D.T. | Methods in Experimental Physics, 9 | 1971 | 23

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This chapter describes the radio wave scattering from the ionosphere. The chapter focuses on the *outdoor* plasmas. Radar scattering measurements made from the ground at frequencies of 50 MHz or higher, frequencies far above the maximum value of the plasma frequency found in the ionosphere are discussed. At such frequencies most of the transmitted energy passes through the ionosphere and is lost; only a small fraction is scattered back to the receiver from small fluctuations in the electron density. As a result, high power transmitters and sensitive receiving equipment and sophisticated data analysis techniques must be used. The chapter discuses incoherent (or Thomson) scatter; it

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involves scattering from random *thermal* fluctuations in the plasma density. The chapter also describes scattering from ion-*acoustic* waves generated by a type of two-stream instability that is encountered in certain regions of the ionosphere. Some general features of all scatter measurements are discussed. Incoherent scatter and the much stronger scattering from unstable regions are examined. The incoherent scatter experiments have shown remarkable quantitative agreement with theory. A very wide range of observations are explained by the linearized kinetic theory. © 1971, Academic Press, Inc.

Article

674 **Generation of Intense Audio Sound Fields Utilizing Arrays of Multiple-Driver Horns** | Levy S.E.,Carlisle R.W. | Journal of the Acoustical Society of America, 33(7) | 1961 | 0

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There are several major applications for intense audio *sound* fields: The *outdoor* propagation of speech and of warning signals, and the sonic fatigue testing of missile and aircraft components. In these applications, several factors combine to make it necessary to limit the portion of the frequency spectrum covered by any one loudspeaker unit. In order to provide adequate power capacity, selection is required between the alternatives of (a) using a large number of identical drivers, (b) using a plurality of drivers each especially designed for a selected portion of the frequency range, and (c) using combinations of these arrangements. In an array currently being used for sonic-fatigue testing, a plurality of large open-cone loudspeakers is used for low frequencies, and a plurality of horns is used for middle and high frequencies. Each horn is driven by several drivers. The driver voice coil and diaphragm proportions have been optimized to withstand fatigue stresses under the *thermal* conditions encountered. The driver is rated at 50-w input for program material. Horn assemblies are available having throat arrays for mounting 6, 12, or 24 drivers. Various applications will be illustrated. © 1961, *Acoustical* Society of America. All rights reserved.

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